This Java program converts an NFA (Non-Deterministic Finite Automaton) into a DFA (Deterministic Finite Automaton) using the Subset Construction method.

Steps :

1. Definition: The program starts by defining the NFA transitions using a HashMap, including the start state (A) and the final state (C).

2. Initialization: The DFA begins with the NFA's start state represented as a set.

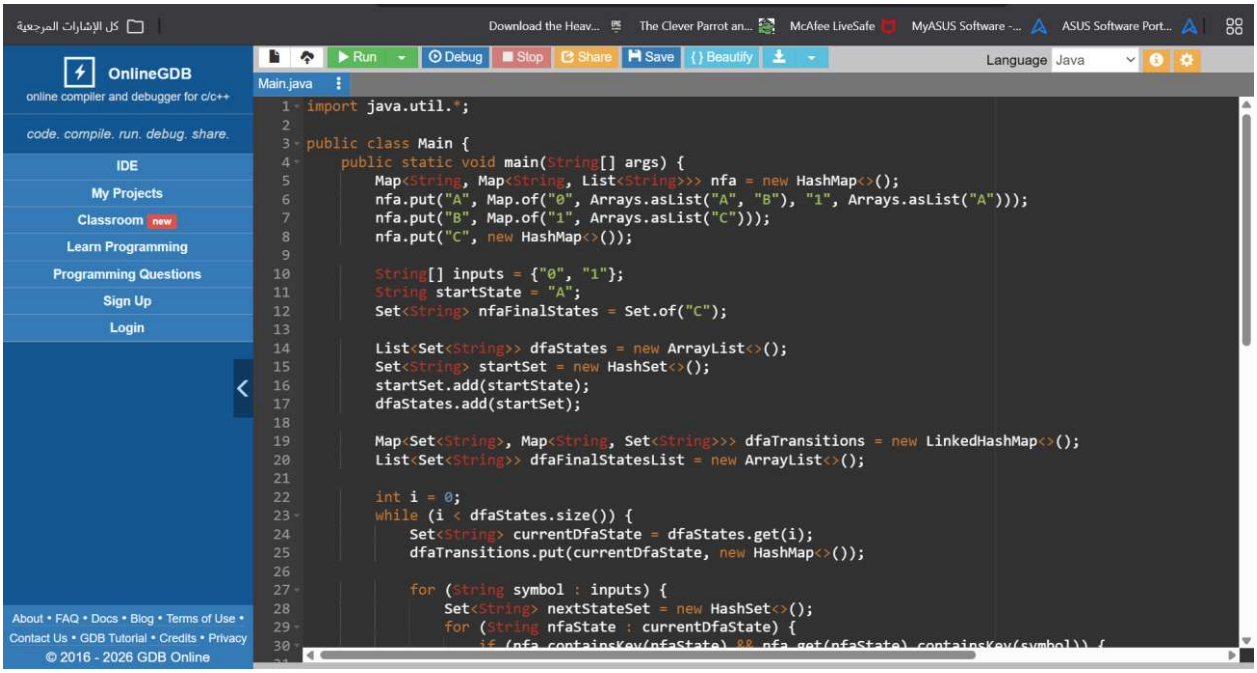
3. Transition Mapping: For each input (0 and 1), the program identifies all reachable NFA states and combines them into a single DFA state.

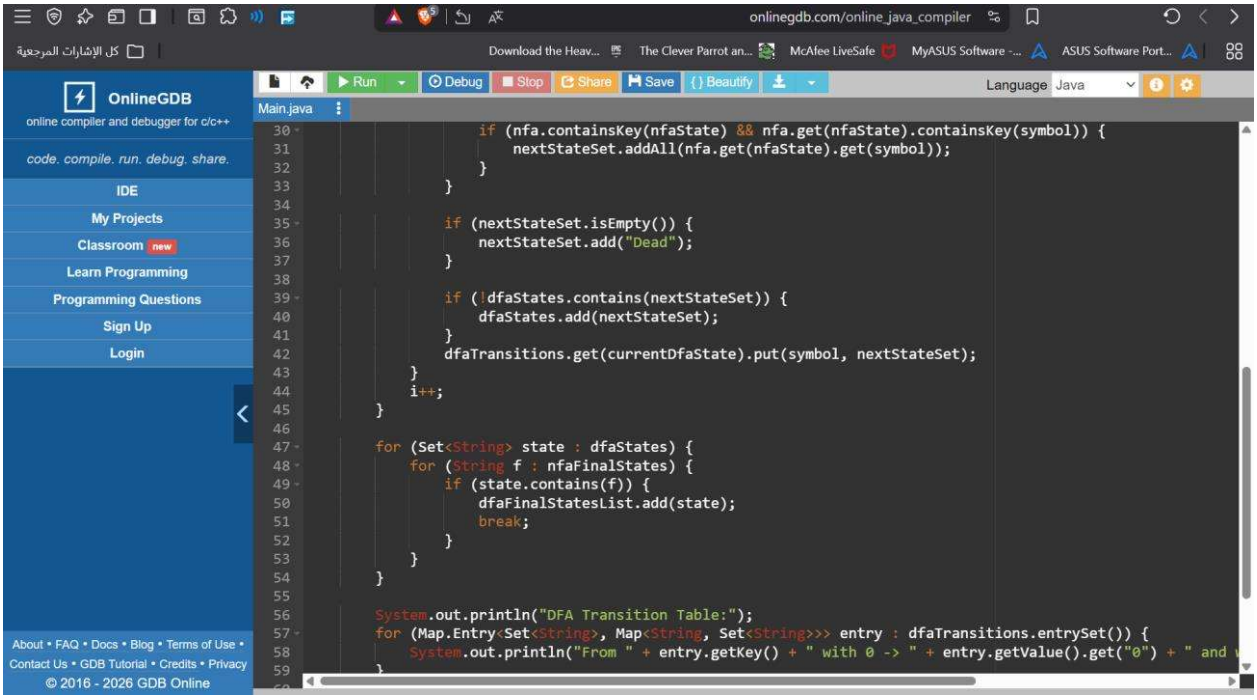
4. Handling Missing Transitions: If no transition exists for a specific input, a "Dead State" is created to maintain determinism.

5. Final State Identification: Any DFA state (set) that contains the original NFA final state (C) is marked as a final state in the DFA.

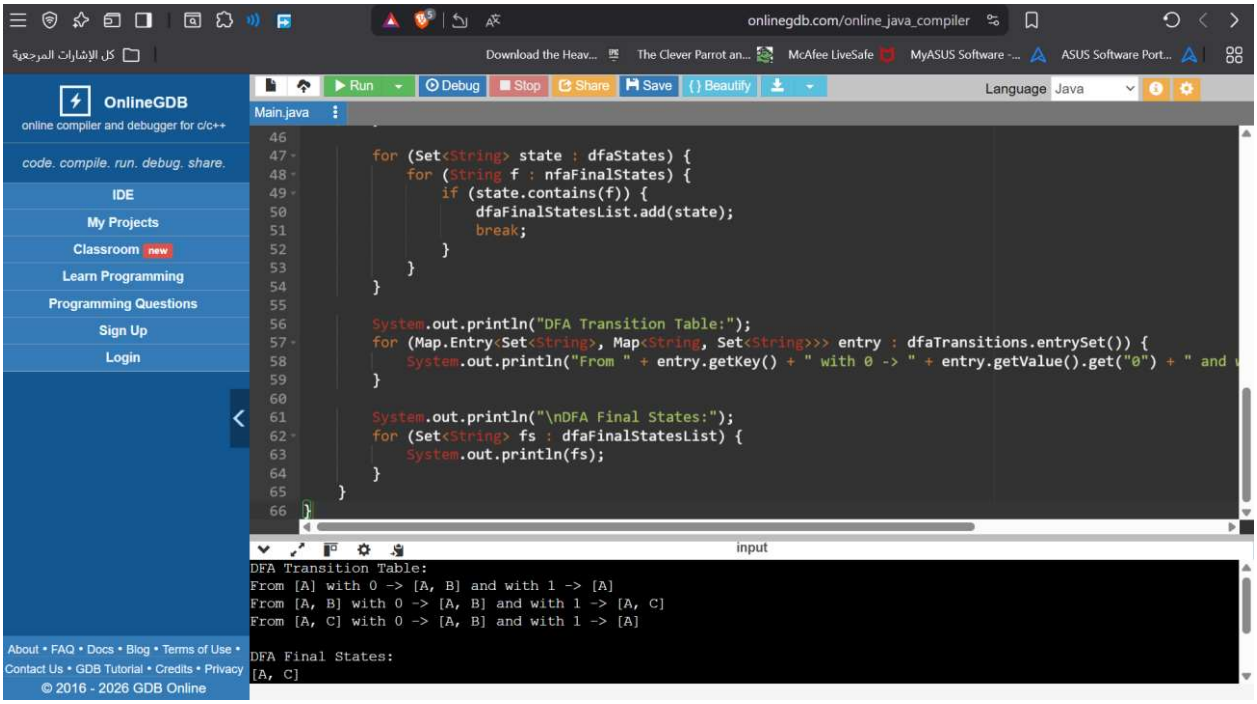
6. Results: The program outputs a clear transition table and lists the final states.

The code:




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Language Java
Main.java
if (nfa.containsKey(nfaState) && nfa.get(nfaState).containsKey(symbol)) {
nextStateSet.addAll(nfa.get(nfaState).get(symbol));
}
}
if (nextStateSet.isEmpty()) {
nextStateSet.add("Dead");
}
if (!dfaStates.contains(nextStateSet)) {
dfaStates.add(nextStateSet);
}
dfaTransitions.get(currentDfaState).put(symbol, nextStateSet);
}
i++;
}
for (Set<String> state : dfaStates) {
for (String f : nfaFinalStates) {
if (state.contains(f)) {
dfaFinalStatesList.add(state);
break;
}
}
}
System.out.println("DFA Transition Table:");
for (Map.Entry<Set<String>, Map<String, Set<String>>> entry : dfaTransitions.entrySet()) {
System.out.println("From " + entry.getKey() + " with 0 -> " + entry.getValue().get("0") + " and
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The Output:


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Language Java
Main.java
for (Set<String> state : dfaStates) {
for (String f : nfaFinalStates) {
if (state.contains(f)) {
dfaFinalStatesList.add(state);
break;
}
}
}
System.out.println("DFA Transition Table:");
for (Map.Entry<Set<String>, Map<String, Set<String>>> entry : dfaTransitions.entrySet()) {
System.out.println("From " + entry.getKey() + " with 0 -> " + entry.getValue().get("0") + " and
}
System.out.println("\nDFA Final States:");
for (Set<String> fs : dfaFinalStatesList) {
System.out.println(fs);
}
}
}
input
DFA Transition Table:
From [A] with 0 -> [A, B] and with 1 -> [A]
From [A, B] with 0 -> [A, B] and with 1 -> [A, C]
From [A, C] with 0 -> [A, B] and with 1 -> [A]
DFA Final States:
[A, C]
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